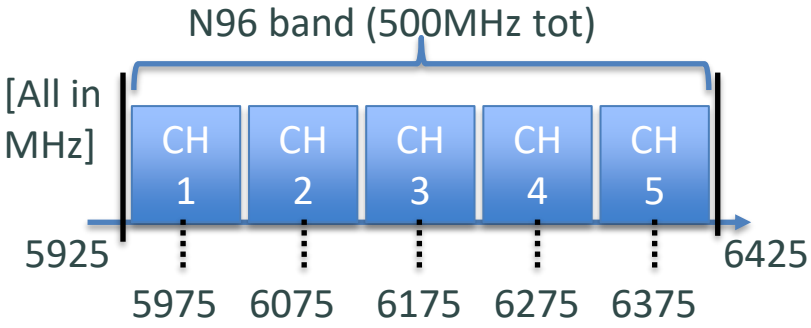
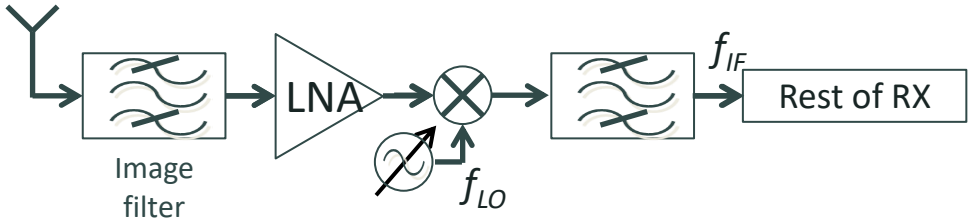
Exercise: RX

A 5G mobile phone must receive in band n96, in any of the 5 standardized 100MHz channels.





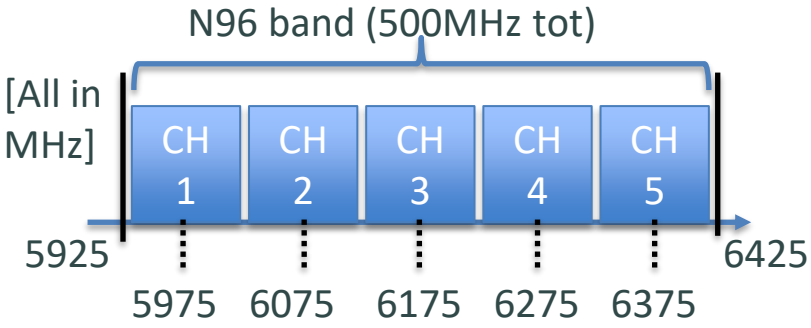
Assume it works at an intermediate frequency f_{IF}=700MHz.

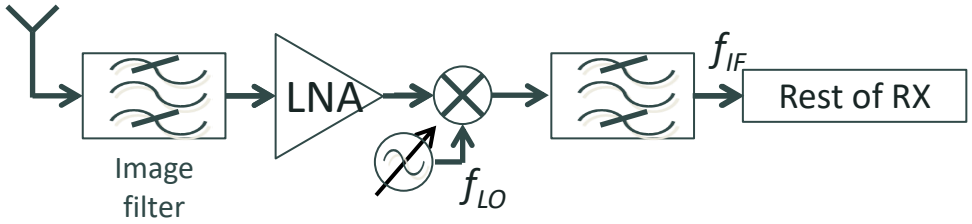
Question 1

- What frequencies should the LO cover? (Choose $f_{LO}<f_{RF}$)
- What is the passband and stopband of the IF filter?
- What is the required passband and stopband for the IM filter?

Question 1a: What frequencies should the LO cover? (Choose $f_{LO}<f_{RF}$)

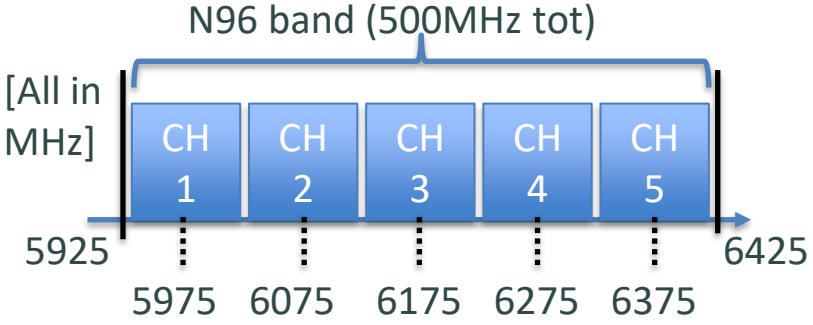
A 5G mobile phone must receive in band n96, in any of the 5 standardized 100MHz channels.





Assume it works at an intermediate frequency f_{IF}=700MHz.

Question 1b: What is the passband and stopband of the IF filter?



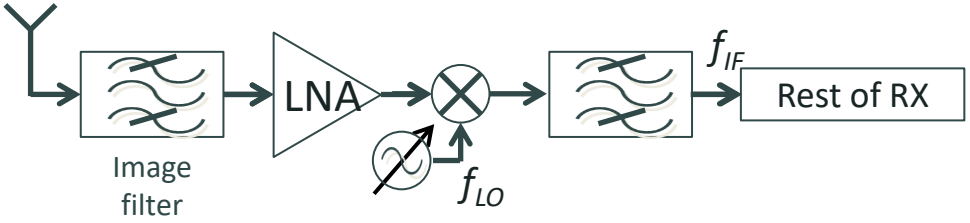
CH1: $f_{LO}=f_1-f_{IF}=5275MHz$

CH2: $f_{LO}=f_2-f_{IF}=5375MHz$

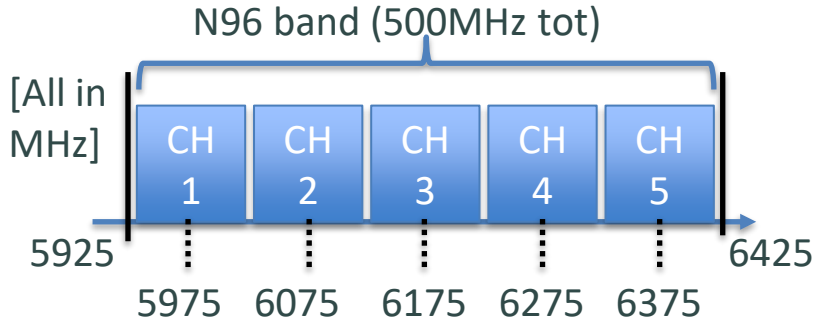
CH3: $f_{LO}=f_3-f_{IF}=5475MHz$

CH4: $f_{LO}=f_4-f_{IF}=5575MHz$

CH5: $f_{LO}=f_5-f_{IF}=5675MHz$



Question 1c: What is the required passband and stopband for the IM filter?



CH1: $f_{LO}=f_1-f_{IF}=5275MHz$

CH2: $f_{LO}=f_2-f_{IF}=5375MHz$

CH3: $f_{LO}=f_3-f_{IF}=5475MHz$

CH4: $f_{LO}=f_4-f_{IF}=5575MHz$

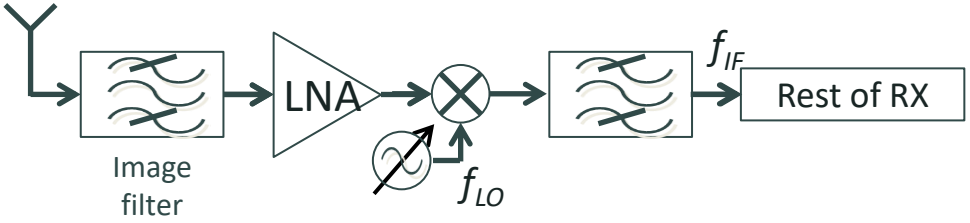
CH5: $f_{LO}=f_5-f_{IF}=5675MHz$

The IF filter is fixed and can be as narrow as one channel (to limit noise/interference from nearby channels):

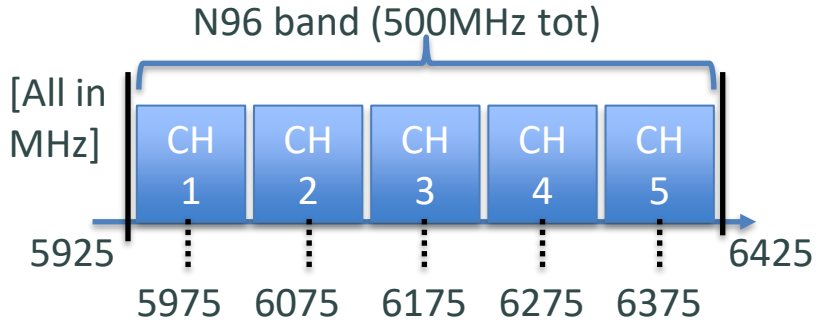
IF filter passband: from 650MHz to 750MHz

The IF filter stopband should at least reject the upper products fLO+fRF, if not already filtered by the mixer or by other components: from 11200 to 12100 MHz

The image band occurs on the other side of the LO



Answer 1a and 1b (LO and IF filter)



CH1: $f_{LO}=f_1-f_{IF}=5275MHz$

CH2: $f_{LO}=f_2-f_{IF}=5375MHz$

CH3: $f_{LO}=f_3-f_{IF}=5475MHz$

CH4: $f_{LO}=f_4-f_{IF}=5575MHz$

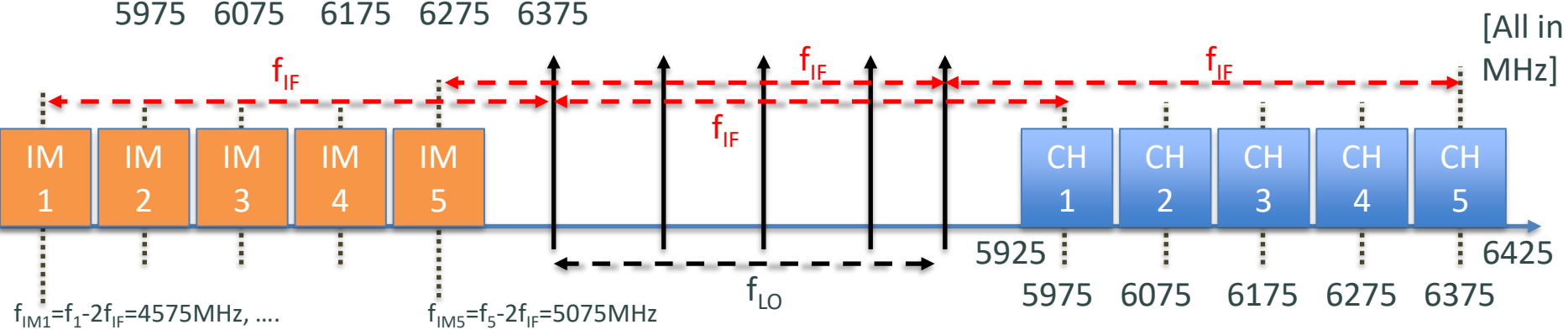
CH5: $f_{LO}=f_5-f_{IF}=5675MHz$

The IF filter is fixed and can be as narrow as one channel (to limit noise/interference from nearby channels):

IF filter passband: from 650MHz to 750MHz

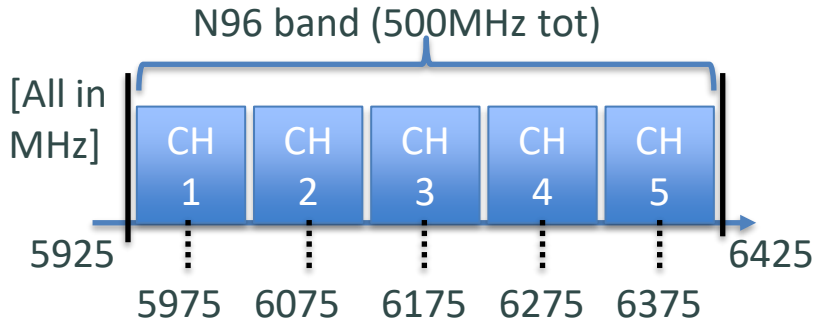
The IF filter stopband should at least reject the upper products fLO+fRF, if not already filtered by the mixer or by other components: from 11200 to 12100 MHz

The image band occurs on the other side of the LO



Each image channel has the same band as the IF filter, which coincides with the useful channel bandwidth of 100MHz

Answer 1c (IM filter)



CH1: $f_{LO}=f_1-f_{IF}=5275MHz$

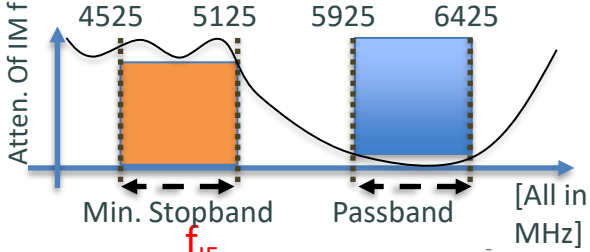
CH2: $f_{LO}=f_2-f_{IF}=5375MHz$

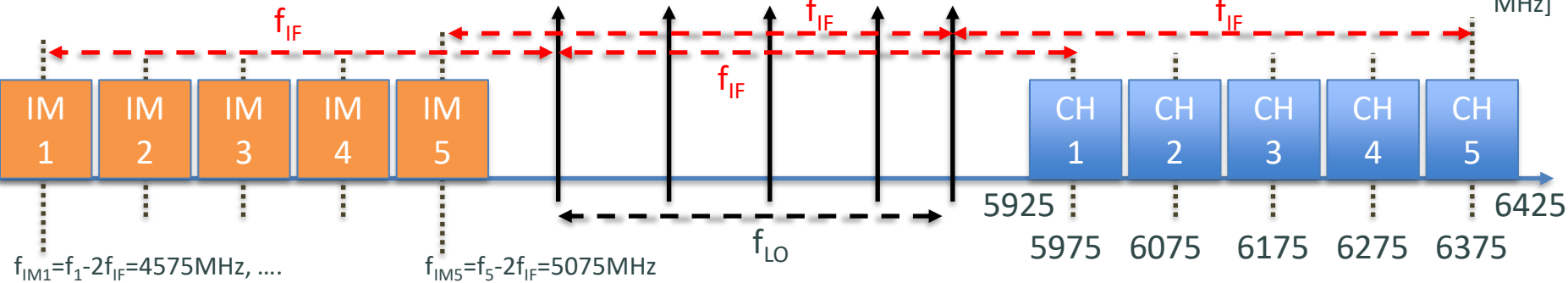
CH3: $f_{LO}=f_3-f_{IF}=5475MHz$

CH4: $f_{LO}=f_4-f_{IF}=5575MHz$

CH5: $f_{LO}=f_5-f_{IF}=5675MHz$

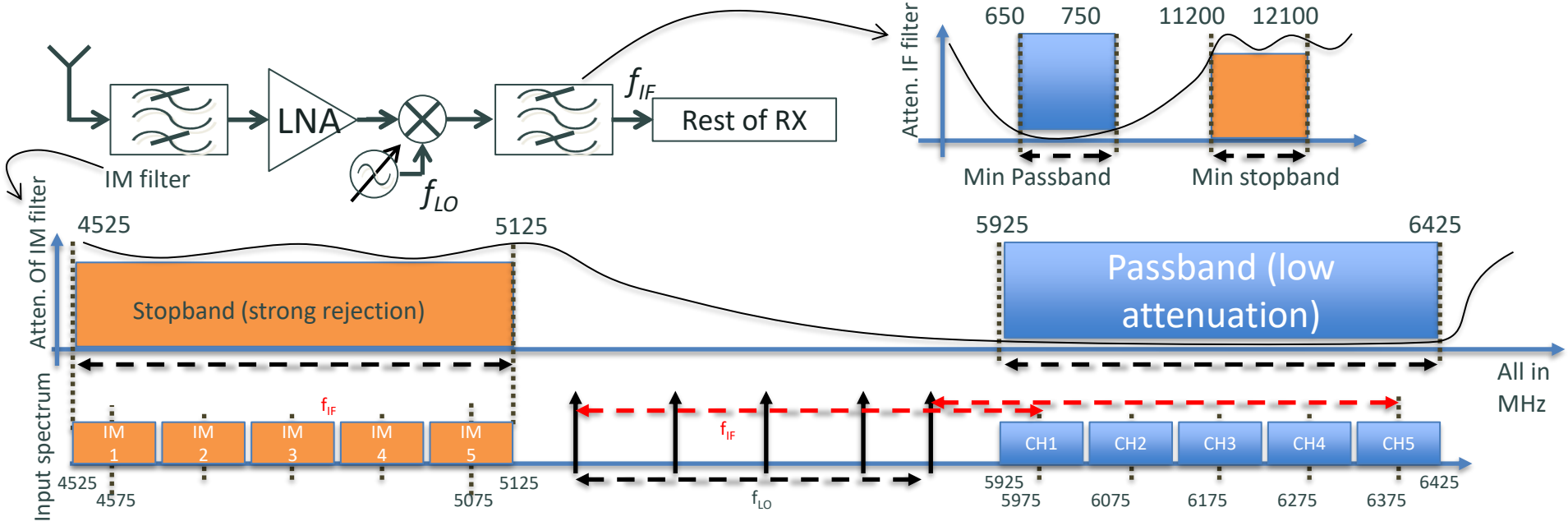
The IM filter is fixed and must let all the useful channels through (passband) and must reject all image channels (stopband)





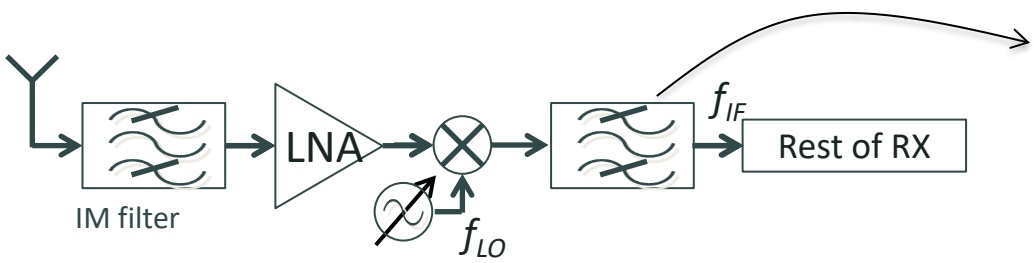
Question 2

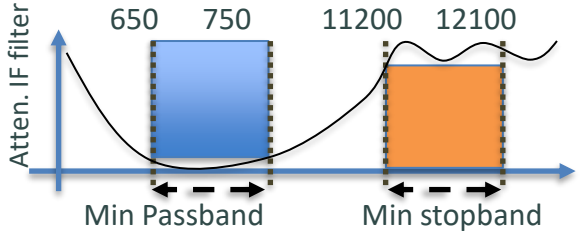
Now we need to receive CH5=-70dBm but **one** among CH1,2,3,4 transmits with strong power. What channel and at what avg power, modeled as 2-tone signal spaced 80MHz, can cause interference at IF as high as the useful signal? Use pure 3rd order non-lin. and IIP3=-20dBm, Gtot=50dB.

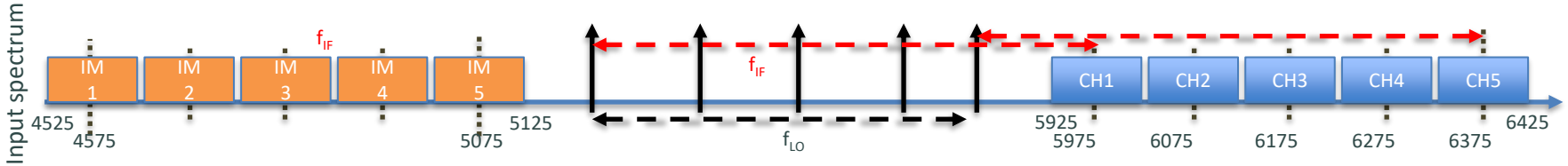


Question 2:

Now we need to receive CH5=-70dBm but **one** among CH1,2,3,4 transmits with strong power. What channel and at what avg power, modeled as 2-tone signal spaced 80MHz, can cause interference at IF as high as the useful signal? Use pure 3rd order non-lin. and IIP3=-20dBm, Gtot=50dB.



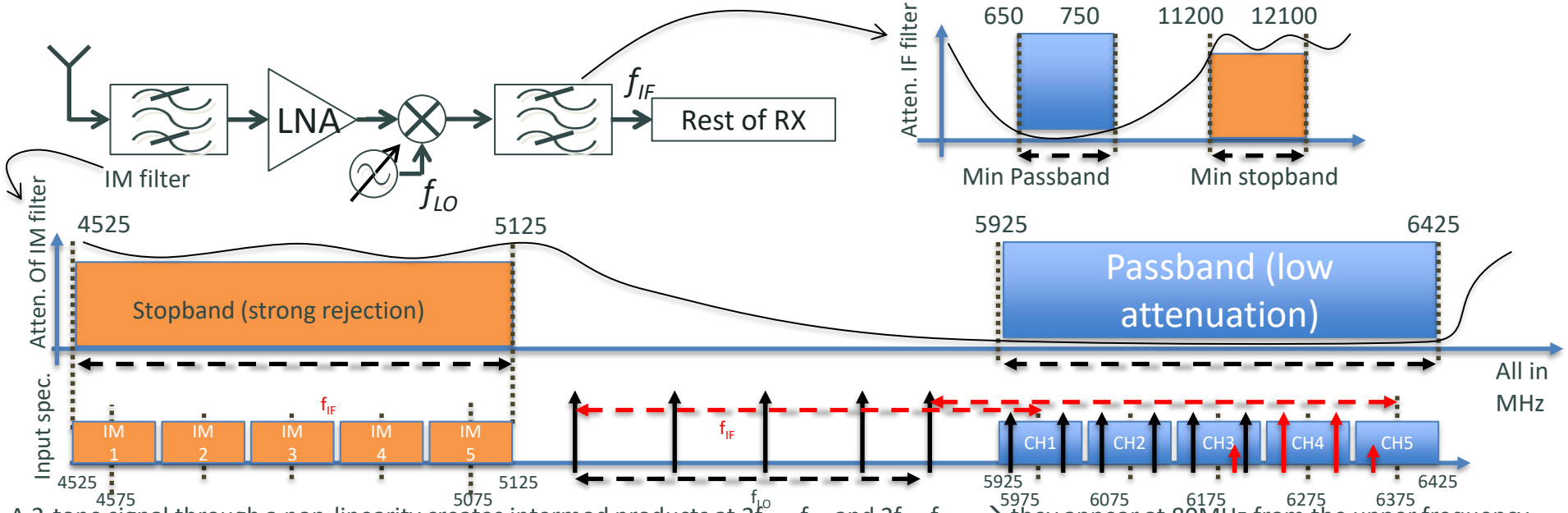




All in MHz

Answer 2: Spectral Regrowth (intermodulation of a channel)

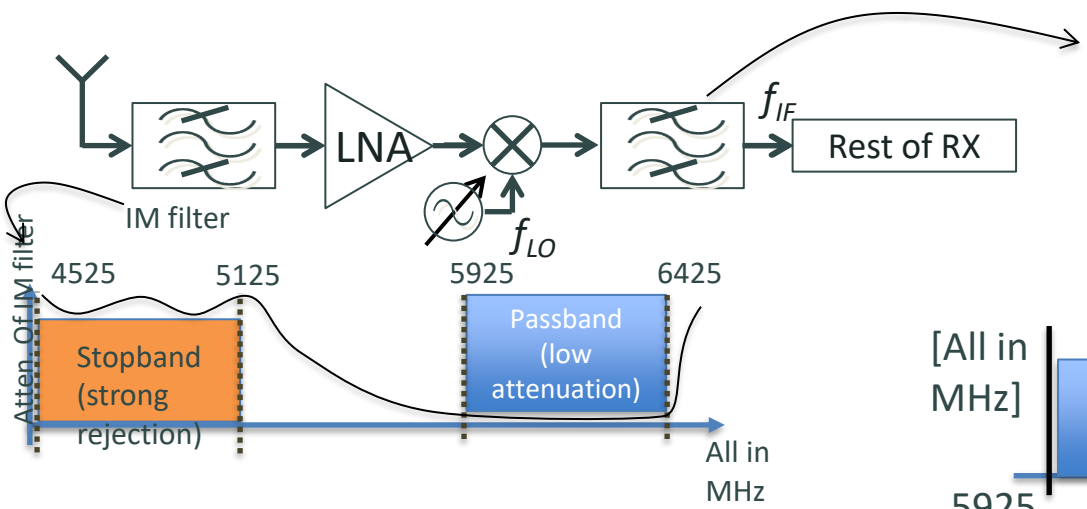
Now we need to receive CH5=-70dBm but one among CH1,2,3,4 transmits with strong power. What channel and at what avg power, modeled as 2-tone signal spaced 80MHz, can cause interference at IF as high as the useful signal? Use pure 3rd order non-lin. and IIP3=-20dBm, Gtot=50dB.

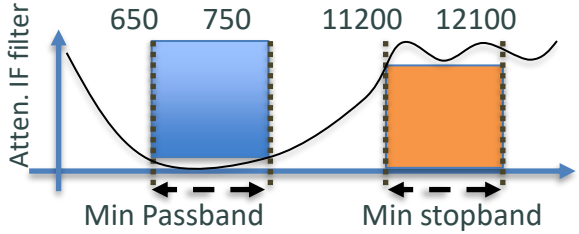


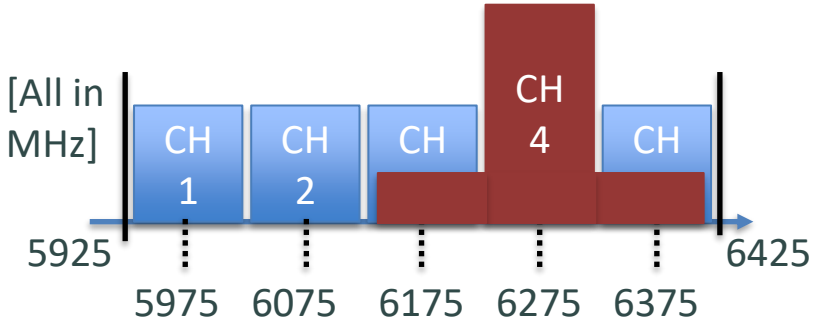
A 2-tone signal through a non-linearity creates intermod products at $2f_{Low}-f_{Up}$ and $2f_{Up}-f_{Low}$. → they appear at 80MHz from the upper frequency and at 80MHz below the lower frequency. → Only CH4 is close enough: one 3rd order IM product falls into CH5 (6275MHz+3*80MHz/2=6395MHz).

$P_{2fUp-fLow}=3P_{fUp}-2\ IP3=3P_{fUp}-2\ IIP3-2G_{tot}$ → To have this equal to -20dBm (useful signal at IF): P_{fUp}=13.3dBm → $P_{CH4@input}=P_{fUp}-G_{tot}+3dB$=-33.7dBm

Comment on Answer 2: Spectral Regrowth (intermodulation of each channel alone)

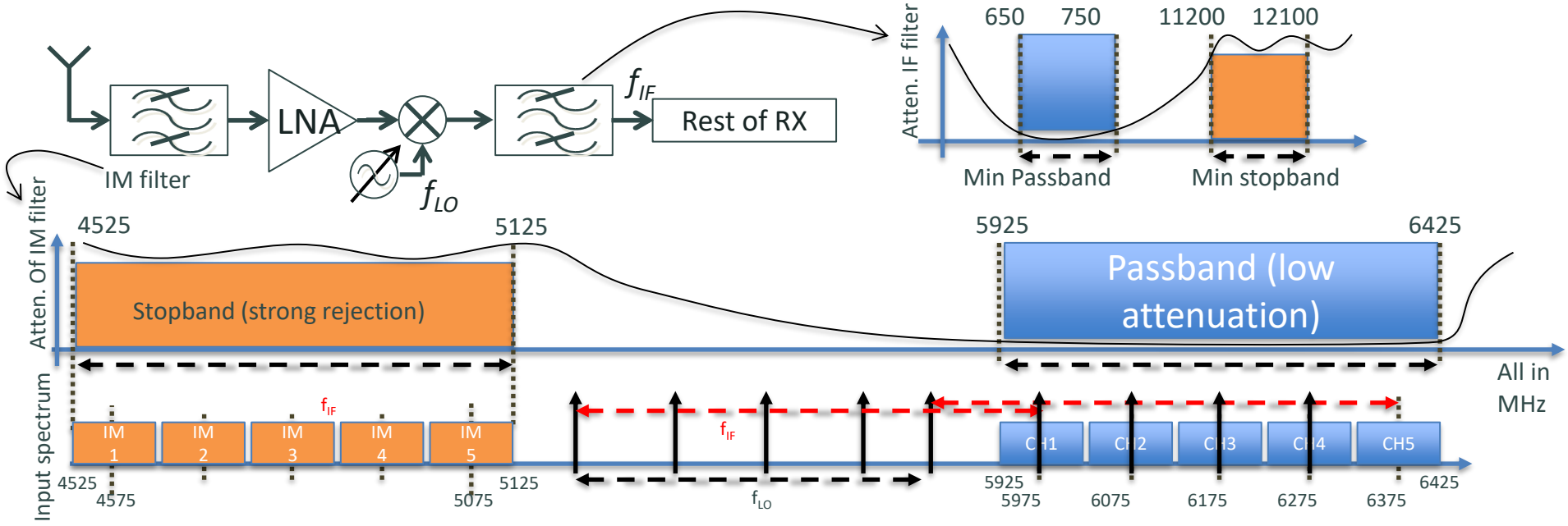






A powerful CH4 through a 3rd order non-linearity is distorted and creates spectral regrowth on its sides, with total bandwidth 3*100MHz=300MHz

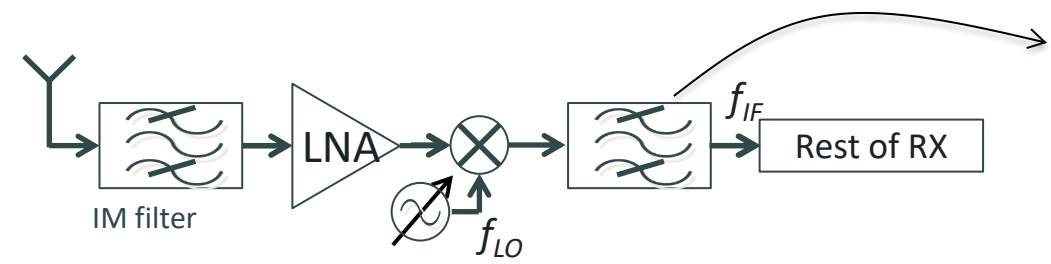
Question 3: Spurious Signals/Intermodulation

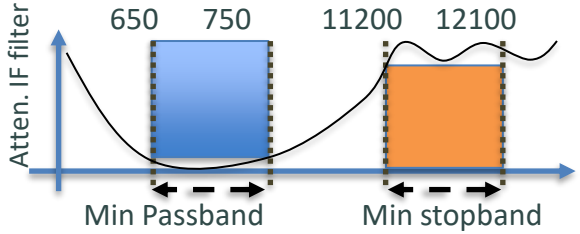


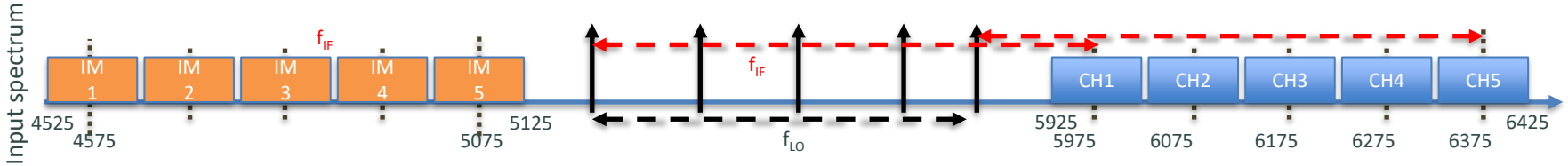
Now we need to receive CH5=-70dBm but **<u>two</u>** among CH1,2,3,4 transmit with strong power. What pair of channels, each modeled as a 1-tone signal, can cause interference at IF? Use pure 3rd order non-lin.

Question 3:

Now we need to receive CH5=-70dBm but **two** among CH1,2,3,4 transmit with strong power. What pair of channels, each modeled as a 1-tone signal, can cause interference at IF? Use pure 3rd order non-lin.



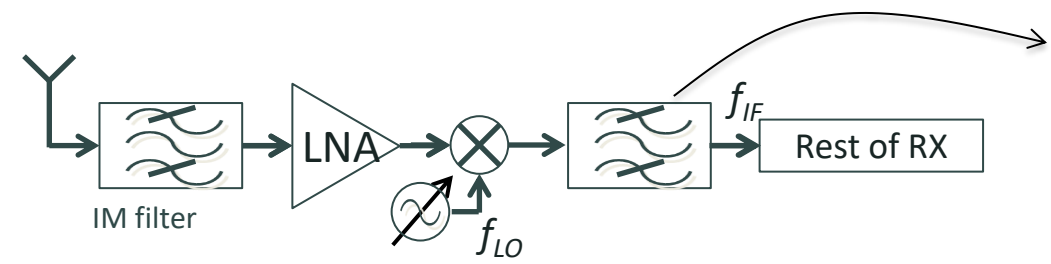


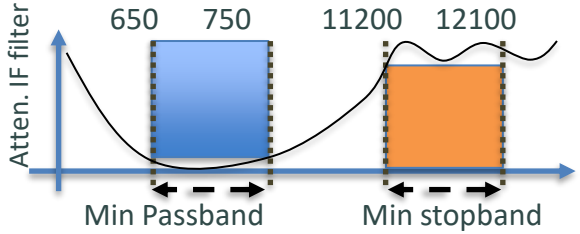


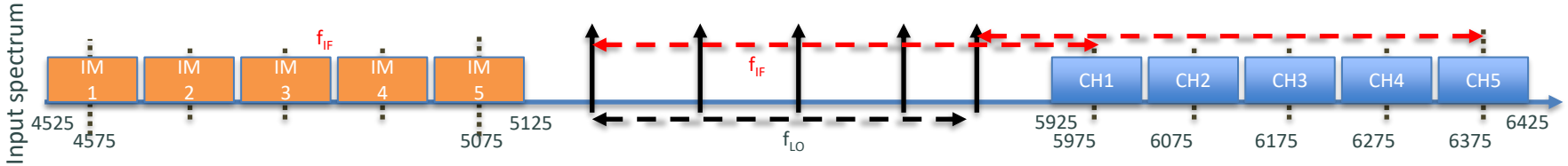
All in MHz

Question 3:

Now we need to receive CH5=-70dBm but **two** among CH1,2,3,4 transmit with strong power. What pair of channels, each modeled as a 1-tone signal, can cause interference at IF? Use pure 3rd order non-lin.







All in MHz

CH3 and CH4 simultaneously turned on constitute a strong 2-tone signal, centered at 6225MHz, spaced 100MHz. Its upper intermod product falls into C5!
CH1 and CH3 simultaneously turned on constitute a strong 2-tone signal, centered at 6075, spaced 200MHz. Its upper intermod product falls into C5!